

Reg. No.

**B.Tech. DEGREE EXAMINATION, NOVEMBER 2019**  
Third Semester

18BTC102J – CELL BIOLOGY

(For the candidates admitted during the academic year 2018-2019 onwards)

**Note:**

- (i) **Part - A** should be answered in OMR sheet within first 45 minutes and OMR sheet should be handed over to hall invigilator at the end of 45<sup>th</sup> minute.
- (ii) **Part - B** and **Part - C** should be answered in answer booklet.

Time: Three Hours

Max. Marks: 100

**PART – A (20 × 1 = 20 Marks)**

Answer **ALL** Questions

1. The prokaryotic cells does NOT contain \_\_\_\_\_.  
(A) Plasma membrane (B) Cell wall  
(C) Cytoplasm (D) Nucleus
2. Piebaldism is an example of defects in \_\_\_\_\_.  
(A) Melanocytes (B) Keratin  
(C) Na<sup>+</sup> (D) Cl<sup>-</sup>
3. Fats are stored in the form of \_\_\_\_\_.  
(A) Fatty acids (B) Triglycerides  
(C) Cholesterol (D) Palmitate
4. Cell junctions are present in \_\_\_\_\_ of plants.  
(A) Vacuoles (B) Plasma membrane  
(C) Cell wall (D) Nucleus
5. Lamins are \_\_\_\_\_ proteins.  
(A) Intermediate filaments (B) Microfilaments  
(C) Microtubules (D) Intermediate tubules
6. Coilin is present in \_\_\_\_\_.  
(A) PML (B) Nuclear speckle  
(C) Cajal bodies (D) Nucleus
7. Lysosomal proteases are active at \_\_\_\_\_ pH.  
(A) Neutral (B) Acidic  
(C) Alkaline (D) Basic
8. Leber's hereditary optic neuropathy is a disease of \_\_\_\_\_ genes.  
(A) Chloroplast (B) Lysosomes  
(C) Nucleus (D) Mitochondria
9. The prokaryotic ancestor of actin is \_\_\_\_\_.  
(A) ABD (B) Mre B  
(C) Fts Z (D) EBS

10. The basic structure of cilia is \_\_\_\_\_  
 (A) Nexin (B) Dynein  
 (C) Axoneme (D) Kinesin
11. Caveolin interacts with \_\_\_\_\_  
 (A) Cholesterol (B) PUFA  
 (C) Fatty acids (D) Triglycerides
12. Life threatening lung disorder is the result of defective \_\_\_\_\_ channel.  
 (A)  $\text{Na}^+$  (B)  $\text{K}^+$   
 (C)  $\text{H}^+$  (D)  $\text{Cl}^-$
13. Cell death expresses \_\_\_\_\_ on cell surface.  
 (A) Phosphatidyl serine (B) Phosphatidyl inositol  
 (C) Phosphatidyl choline (D) Phosphatidyl ethanolamine
14. Adenyl cyclase is responsible for \_\_\_\_\_ formation.  
 (A) cGMP (B) cUMP  
 (C) cAMP (D) cCMP
15.  $\text{IP}_3$  acts to release \_\_\_\_\_ from ER.  
 (A)  $\text{Ca}^{2+}$  (B)  $\text{K}^+$   
 (C)  $\text{Mg}^{2+}$  (D)  $\text{Na}^+$
16. Stem cells give rise to all type of cells are called \_\_\_\_\_.  
 (A) Omnipotent (B) Pleuripotent  
 (C) Multipotent (D) Unipotent
17. Solid tumors in connective tissues are called \_\_\_\_\_.  
 (A) Carcinoma (B) Sarcoma  
 (C) Lymphoma (D) Leukemia
18. Herceptin is used for the treatment of \_\_\_\_\_ cancer.  
 (A) Breast (B) Oral  
 (C) Cervical (D) Lung
19. Tangles of neurodegenerative disorder contains \_\_\_\_\_ protein.  
 (A) Ras (B) Raf  
 (C) ERK (D) Tau
20. Genes from which retroviral oncogenes originated are called \_\_\_\_\_.  
 (A) Autocrine growth stimulation (B) Avian cells  
 (C) Proto oncogenes (D) EGF receptor

**PART – B (5 × 4 = 20 Marks)**  
 Answer ANY FIVE Questions

21. Protein molecules chemical evolution is the cause of origin of universe. Explain.
22. Describe the structure that separates the nucleus from cytoplasm.

23. Explain the hypothesis of vesicle docking and fusion from Golgi complex.
24. Differentiate symport, uniport and antiport with examples.
25. Illustrate the protein mechanism in different phases of cell cycle.
26. Write a note on necroptosis.
27. Describe the genes that mutated during cancer formation.

**PART – C (5 × 12 = 60 Marks)**  
 Answer ALL Questions

28. a. Describe the molecular composition of cells.  
 (OR)  
 b. Describe the structure, composition and function of semipermeable membrane in cells.
29. a. Write a brief note on protein sorting and folding in ER.  
 (OR)  
 b. ATP producing structures has a main role in energy metabolism. Justify.
30. a. Explain the mechanism of muscle contraction.  
 (OR)  
 b. Smaller molecules can be transported across cell membrane without energy. Explain with diagram.
31. a. Adenyl cyclase catalyzes the ATP conversion to second messenger. Explain the mechanism with diagram.  
 (OR)  
 b. Write a note on stem cells and its applications.
32. a. Discuss about the tumors caused by HPV and causes leukoplakia.  
 (OR)  
 b. Discuss about the disorders of neurons.

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